Extending Bulk-Billing Incentives to All Patients: Evidence from Australia's November 2025 Medicare Reform

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Abstract

On 1 November 2025 Australia extended Medicare bulk-billing incentives, previously limited to children and concession card holders, to all patients, and introduced a 12.5% loading for practices that bulk bill every eligible service. Comparing working-age adults, who became eligible, with children, who already were, in monthly Medicare data for 393 local government areas by age group, I find that the reform raised the working-age bulk-billing rate by 6.3 percentage points (95% confidence interval 5.7 to 6.9) from a base of 69%. The effect appeared in the first month and was still growing eight months later, and it closely matches the estimate of Craig et al. (2026), obtained from the same data with a different design. The response was largest in areas where the fewest working-age bulk-billed attendances already attracted the incentive. The paper's main contribution is the cost accounting. Because incentive payments are observed by age group, the incentives already paid for working-age concession card holders before the reform can be netted out. I show that the cost per additional bulk-billed attendance depends on the effect, the counterfactual bulk-billing rate and the share of attendances already eligible, and that assuming no prior eligibility raises the cost by about half. The incentive cost of an additional bulk-billed attendance is about $187 ($171 to $207), against $322 in Craig et al. (2026), and 88% of the additional incentive spending went to attendances that would have been bulk billed anyway. I also show that estimates of the November 2023 tripling from the same age-group contrast depend heavily on the pre-period: a six-month window reproduces the 3-point estimate of Craig et al. (2026), while longer windows show two years of divergence beforehand. A sensitivity analysis following Rambachan and Roth (2023) shows that the jump in November 2023 is robust, but the average effect over the following two years is not.

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1 Introduction

Bulk billing, where a general practitioner (GP) accepts the Medicare benefit as full payment so that the patient pays nothing at the point of service, is the main way Australia keeps primary care free for most patients. The share of GP attendances that were bulk billed fell sharply after the COVID-19 period, and the government responded with two large increases in the bulk-billing incentive, an extra payment made to GPs when an eligible patient is bulk billed. In November 2023 the incentive was tripled for children and concession card holders. In February 2025 the government announced an $8.5 billion package that, from 1 November 2025, extended the incentive to every Medicare-eligible patient and added a Bulk Billing Practice Incentive Program (BBPIP) paying a 12.5% loading to practices that bulk bill all eligible services, with a stated aim of nine in ten GP visits being bulk billed by 2030 (Prime Minister of Australia, 2025).

Conditional subsidies of this kind are expensive when compliance is already high, because they are paid on every eligible service, including services that would have been provided on the same terms anyway. Craig et al. (2026) make this point forcefully for the 2023 and 2025 incentive changes, estimating that each additional bulk-billed appointment caused by the 2025 reform cost the government $322 in incentive payments. This paper re-examines the 2025 reform with different data and a different design, and makes three contributions.

First, it provides an independent estimate of the reform's effect. Comparing working-age adults, who became eligible, with children, who were already eligible, in monthly data by local government area (LGA) and age group from November 2023 to June 2026, I find that the reform raised the working-age bulk-billing rate by 6.3 percentage points (95% confidence interval 5.7 to 6.9) from a base of 69%. Most of the effect appeared in November 2025, and it was still rising in June 2026. Because the comparison is made within each LGA and month, the estimate is robust to any national or local shock in November 2025 that affected both groups, which a before-and-after design for each age group cannot rule out. The estimate agrees closely with Craig et al. (2026), who use the same LGA data and whose estimates for 16–64 and 0–15 year olds imply a difference of 6.1 points. The response was about twice as large in the LGAs where the fewest working-age bulk-billed attendances already attracted the incentive as in those where the most did.

Second, and most importantly, it corrects the cost accounting. Working-age concession card holders were eligible for the incentive before November 2025, so part of the incentive paid on working-age attendances after the reform is not new spending. Craig et al. (2026) cannot observe concession status in their data and assume that no 16–64 year old was previously eligible, which they note leads them to overstate the cost. Incentive payments by age group are observed in the data used here, so the incentives already being paid can be netted out. I show in Section 6 that the cost per additional bulk-

billed attendance can be written in terms of four observable quantities: the incentive per bulk-billed attendance, the effect, the counterfactual bulk-billing rate, and the share of bulk-billed attendances that already attracted the incentive. Before the reform about 35% of working-age bulk-billed attendances did. Netting this out lowers the incentive cost per additional bulk-billed attendance from $276 to $187 on my estimates ($171 to $207 across the confidence interval). The same formula, applied to their inputs, reproduces their $322. The qualitative conclusion is unchanged: about 88% of the additional incentive spending went to attendances that would have been bulk billed anyway.

Third, it shows that estimates of the November 2023 tripling from the same kind of age-group contrast depend heavily on the pre-period. In the state-level data the gap between children's and working-age adults' bulk-billing rates widened from about 7 points in mid-2021 to 18 points in mid-2023, before the tripling. With a six-month pre-period, as in Craig et al. (2026), I reproduce their estimate of about 3 points, and trends over that window look roughly parallel in log-odds, though not in percentage points. With longer pre-periods the estimate rises to 8 to 10 points and the pre-trends are clearly not parallel. Their six-month window was chosen because trends over it looked parallel, and it begins in the month the tripling was announced. Following Rambachan and Roth (2023), I show that the jump in bulk billing in November 2023 survives large departures from parallel trends. The average effect over the following two years does not: it loses significance under changes in the slope of the differential trend smaller than the change observed in the year before the tripling, and in percentage points it reverses sign if the pre-period trend is extended. The size of the 2023 effect, and the cost per additional bulk-billed attendance that rests on it, are therefore not well pinned down by aggregate data.

One caveat applies to the design. The model in Craig et al. (2026) predicts that raising the return to bulk billing one group makes GPs' time more valuable and so makes already-eligible groups weakly less likely to be bulk billed. If the 2025 reform reduced bulk billing of children in this way, the comparison with children overstates the effect on working-age adults and understates the cost per additional attendance. Children's bulk-billing rate in fact rose slightly after the reform, so any such spillover is outweighed by the practice loading and cannot be measured separately here; I discuss its likely size in Section 5.4.

1.1 Related literature

The paper contributes to work on conditional subsidies to providers. Payments conditional on provider behaviour have raised the use of targeted services in low-income settings (Morris et al., 2004; Basinga et al., 2011), while pay-for-performance in high-income primary care has produced modest gains that fade once compliance is high (Rosenthal et al., 2005; Campbell et al., 2009). Bulk-billing incentives are a clean case of a subsidy conditional on a pricing decision, paid on every eligible service whether or not it would have

been bulk billed anyway, so the infra-marginal share of spending is large when baseline compliance is high. Craig et al. (2026) formalise this in a model of GP pricing with limited eligibility. The accounting in Section 6 makes explicit how the infra-marginal share depends on prior eligibility as well as on baseline compliance.

Evidence on bulk billing in Australia has been largely descriptive. De Abreu Lourenço et al. (2015) find that GPs bulk bill low-income patients more often; Graham et al. (2023) find no consistent spatial association between bulk-billing-only practices and area socioeconomic status; Mudiyanselage et al. (2023) associate the Medicare rebate freeze with lower bulk billing; O'Sullivan et al. (2022) discuss the rural effects of mandatory bulk billing during COVID-19; and Elkins and Schurer (2017) model the incidence of a proposed GP co-payment. Saxby and Zhang (2025a) document bulk-billing rates and out-of-pocket costs by region, and Saxby and Zhang (2025b) estimate the effect of the 2022 rural incentive loading with a difference-in-differences design. On the supply side, Australian GPs' labour supply and location choices respond only weakly to financial incentives (Broadway et al., 2017; Yong et al., 2018; McIsaac et al., 2019), and GP prices respond to local competition (Gravelle et al., 2016). Findlay (2026) and Craig et al. (2026) provide the first causal estimates for the 2022, 2023 and 2025 incentive changes.

The remainder of the paper is organised as follows. Section 2 describes the policies, Section 3 the data and Section 4 the empirical strategy. Section 5 reports the estimates, Section 6 the cost accounting, Section 7 the comparison with Craig et al. (2026) and Section 8 the 2023 tripling. Section 9 concludes.

2 Institutional background

Medicare pays a benefit, usually 85% of the Medicare Benefits Schedule (MBS) fee for out-of-hospital services, for each eligible GP attendance.[1] A GP may charge the patient more than the benefit, or may bulk bill, accepting the benefit as full payment. When an eligible patient is bulk billed, the GP also receives the bulk-billing incentive, a separate MBS item whose amount rises with remoteness under the Modified Monash (MM) classification. Before November 2025, eligibility for the incentive was limited to children under 16 and holders of Commonwealth concession cards, a group that includes most people aged 65 and over and a minority of working-age adults.

November 2023: tripled incentive. From 1 November 2023 new incentive items paid three times the standard amount for eligible patients receiving face-to-face general attendances longer than six minutes (Levels B to E) and longer video and telephone consultations for patients registered through MyMedicare (The Hon Mark Butler MP,

[1] All incentive amounts in this paper are benefits (85% of the schedule fee), taken from the MBS XML schedule files (Department of Health, Disability and Ageing, 2025).

2023). Eligibility did not change. In metropolitan areas (MM1) the incentive rose from \$6.85 to \$20.65 per eligible service, and in very remote areas (MM7) from \$13.15 to \$39.70.

November 2025: all patients and the practice loading. The package announced on 23 February 2025 and taken to the May 2025 federal election made two changes from 1 November 2025 (Prime Minister of Australia, 2025). First, the incentive became payable for all Medicare-eligible patients. Second, the BBPIP added a quarterly payment equal to 12.5% of MBS benefits for eligible bulk-billed services, split equally between practice and GP, for practices that bulk bill all eligible services and are registered through MyMedicare. At the July 2025 schedule the tripled incentive was \$21.85 in MM1 and \$42.05 in MM7 (Table 8).

The design exploits the first change. Children were fully eligible before the reform, so for them the reform changed only the practice loading; the incentive paid per bulk-billed child attendance rose by 2%, from \$22.93 to \$23.46, which reflects indexation. Working-age adults became eligible, and the incentive paid per bulk-billed working-age attendance rose from \$7.96 to \$22.73 (Table 1). Seniors are in between: most were already eligible as concession card holders, but the incentive paid per bulk-billed senior attendance rose by 25%, from \$18.22 to \$22.79, so seniors were partly treated.

3 Data

The main data are the Australian Institute of Health and Welfare's monthly series on Medicare bulk billing of GP attendances (Australian Institute of Health and Welfare, 2026). Supplementary Table 3 reports, for each local government area (LGA) and each of three age groups (0–15, 16–64, 65+), the bulk-billing rate, the number of bulk-billed GP attendances, bulk-billed GP attendance benefits and bulk-billing incentive benefits. Table 2 reports the same series by state and territory. Both run monthly to June 2026, and cells with few attendances are suppressed in the LGA table. The analysis panel covers November 2023 to June 2026 and keeps the 393 LGAs in which all three age groups are reported in every month. Together they account for 99.5% of GP attendances in the published LGA data. The panel has 393 LGAs $\times$ 3 age groups $\times$ 32 months, or 37,728 observations, with 24 pre-reform months and 8 post-reform months. Starting in November 2023 keeps the 2023 tripling out of the pre-period. The summary statistics and the national totals used in the cost accounting are taken from the state table, which has no suppression; the published LGA cells sum to 98% of them.

Total GP attendances are not reported directly. I derive them as bulk-billed attendances divided by the bulk-billing rate. Because the rate is published to 0.1 percentage point, derived totals carry a rounding error of around 0.1 to 0.2%. Summed to quarters, the derived national totals match the official quarterly counts of GP non-referred atten-

dances in the Medicare statistics by Modified Monash category (Department of Health, Disability and Ageing, 2026) to within 2% in every quarter since 2018–19, and within 1% in most. Reported attendance counts are fractional, which suggests some scaling in the published series; I use them as published. The rounding error in derived totals is larger in small LGAs, which is one more reason to weight cells by attendances.

The incentive benefits reported by age group are what make the cost accounting in Section 6 possible: they show directly how much incentive was being paid on working-age attendances before the reform, without needing to observe who held a concession card.

Two further sources are used descriptively. The quarterly Medicare statistics by Modified Monash category give bulk-billing rates for all ages by remoteness zone, and the MBS XML schedule files give incentive amounts by zone (Department of Health, Disability and Ageing, 2026, 2025). Neither has an age split, so neither enters the main estimates.

Table 1 summarises the panel. Before the reform, 69% of working-age GP attendances were bulk billed, against 90% for children and 87% for seniors. Working-age adults account for about 7.5 million GP attendances a month, more than children and seniors combined. Incentive payments for working-age adults more than tripled, from $40 million to $130 million a month.

4 Empirical strategy

4.1 Difference-in-differences

The main specification is

$$y_{agt} = \beta \left(W_g \times P_t \right) + \alpha_{ag} + \delta_{at} + \varepsilon_{agt}, \tag{1}$$

where y_{agt} is the bulk-billing rate (in percentage points) of age group g in LGA a in month t, W_g indicates working-age adults, P_t indicates months from November 2025, α_{ag} are LGA-by-age-group fixed effects and δ_{at} are LGA-by-month fixed effects. The LGA-by-month effects absorb seasonality and any shock common to all age groups in an area, whether national or local, including the common component of the practice loading. The effect is therefore identified only from the gap between age groups within the same LGA and month. I also report estimates with month effects in place of LGA-by-month effects. In the preferred specification the comparison group is children; I also report specifications using seniors and children and seniors pooled. Cells are weighted by their mean monthly GP attendances before the reform, so β is an attendance-weighted average effect, the quantity needed to convert the estimate into national attendances and costs (Solon et al., 2015); Craig et al. (2026) likewise weight by appointments. Unweighted estimates, which average across LGAs, are reported in Table 3.

The identifying assumption is that, absent the reform, the working-age bulk-billing rate would have moved in parallel with the comparison group. Because the reform is national and the treated group is one age band, a separate shock affecting working-age adults alone in November 2025 would be indistinguishable from the reform. I examine this with an event study, placebo reform dates within the pre-period, a placebo comparison of seniors against children, and a test for differential linear trends before the reform.

Inference. Standard errors are clustered by LGA, which allows for arbitrary correlation over time and across age groups within an area and gives 393 clusters. Shocks may also be correlated across LGAs within a state, for example through state health policy, so I also report standard errors clustered by state, with confidence intervals from $t(7)$ critical values. These are about twice as large but leave every conclusion unchanged. With only eight state clusters, conventional cluster-robust standard errors over-reject (Cameron et al., 2008), so the state-level estimates in Appendix Table 9 use the restricted wild cluster bootstrap with Webb weights (Webb, 2023; Roodman et al., 2019), which is conservative when clusters are as unequal as Australian states (MacKinnon and Webb, 2017).

4.2 Event study

The event-study specification replaces $W_g \times P_t$ in equation (1) with interactions of W_g with each month, omitting October 2025:

$$y_{agt} = \sum_{k \neq 0} \beta_k \, W_g \cdot \mathbf{1}[t = k] + \alpha_{ag} + \delta_{at} + \varepsilon_{agt}. \tag{2}$$

Pre-reform coefficients test for anticipation and differential trends; post-reform coefficients trace the dynamics of the response.

5 Results

5.1 Main estimates

Figure 1 shows national bulk-billing rates by age group. The working-age rate was flat at around 69% from late 2023 to October 2025, jumped from 69.7% in October 2025 to 75.1% in November, and reached 78.3% in June 2026. The rates for children and seniors rose by much less.

Table 2 reports estimates of equation (1). In the preferred specification (column 1), the reform raised the working-age bulk-billing rate by 6.33 percentage points relative to children (standard error 0.31; 95% confidence interval 5.7 to 6.9). Clustering by state widens the interval to 4.8 to 7.9. Using seniors as the comparison group gives 6.16 points, and pooling children and seniors gives 6.20 points. The slightly smaller estimate against

seniors is expected, because seniors were partly treated. Replacing the LGA-by-month effects with month effects (column 4) gives 6.40 points, so local shocks play almost no role. The same comparison on the state-level panel gives 6.43 points (Appendix Table 9).

5.2 Dynamics

Figure 2 plots the event-study coefficients, relative to children (panel a) and to children and seniors pooled (panel b). Against children, the coefficients from November 2023 to June 2025 average −0.2 points with no clear pattern, while those for July to September 2025 are lower (−1.3, −0.6 and −0.5 points), so October 2025 was a slightly high month for working-age adults. The pooled coefficients sit at about −1.0 points throughout the pre-period. The effect appears immediately, at 5.0 points in November 2025 relative to October against children (4.4 pooled), and rises to 6.9 points by June 2026 (6.1 pooled). With only eight post-reform months, the long-run effect is not yet known.

5.3 Robustness

Table 3 collects the robustness checks. Dropping one state at a time moves the estimate between 6.01 (Victoria excluded) and 7.08 points (New South Wales excluded). Excluding unincorporated areas leaves it unchanged, and the effect in log-odds is 0.30. The unweighted estimate, which averages across LGAs rather than attendances, is larger at 8.1 points, because smaller LGAs responded more. Because the cost accounting applies the effect to national attendances, the attendance-weighted estimate is the relevant one.

With 393 clusters the placebo and pre-trend tests are more demanding, and some small differences that could not be distinguished from zero in the state data are now precisely estimated. Before the reform the working-age rate drifted down relative to children by 0.034 points a month and up relative to seniors by 0.027 points a month. Both are statistically significant but small. Because the post-reform months are on average 16 months later than the pre-reform months, trends of this size would account for between about −0.5 and +0.4 points of a difference-in-differences estimate. The trend against children runs against finding an effect, so extrapolating it would raise the preferred estimate slightly: if anything, the preferred estimate is conservative. Placebo reform dates against children give negative estimates that shrink towards the reform: −0.77 points for May 2024, −0.35 for November 2024 and −0.17 for May 2025 ($p = 0.05$). A linear trend of −0.034 points a month implies about −0.4 points at each date, so the drift against children appears concentrated in 2024 and had largely flattened in the year before the reform. Against children and seniors pooled, the May and November 2024 placebos are close to zero, and the May 2025 placebo is 0.46 points ($p < 0.001$). This could reflect anticipation after the February 2025 announcement and the May 2025 election, but it is under a tenth of the main effect.

The placebo comparison of seniors against children gives +0.27 points ($p = 0.004$). A small positive effect for seniors is consistent with their partial treatment: non-concession seniors became eligible, and the incentive paid per bulk-billed senior attendance rose by 25%. Craig et al. (2026) likewise find a larger effect for 65+ than for 0–15 year olds (1.18 against 0.42 points).

Exposure. If the reform works through eligibility, areas in which more working-age bulk-billed attendances newly attracted the incentive might be expected to respond more. I measure an LGA's exposure as one minus the ratio of pre-reform (November 2024 to October 2025) to post-reform incentive paid per bulk-billed working-age attendance. Between the 10th and 90th percentiles of LGAs it ranges from 0.46 to 0.76. The response falls with exposure rather than rising (Figure 5): the effect is 10.6 points in the fifth of LGAs with the lowest exposure and 4.7 points in the fifth with the highest. Interacting $W_g \times P_t$ with exposure gives −16.5 points per unit of exposure (standard error 2.5). The coefficient is unchanged when an interaction with the pre-reform working-age bulk-billing rate is added, and the baseline rate itself does not predict the effect (−0.03 points per 10 points, standard error 0.25). The same pattern was visible across the eight states, where it could not be distinguished from noise.

Exposure is measured among bulk-billed attendances, which limits its interpretation. Low exposure means that most working-age attendances bulk billed before the reform were those of concession card holders, which suggests that other working-age patients were rarely bulk billed in those areas and so had the most room to respond. The result is therefore better read as showing that the reform had most effect where non-concession adults had been least likely to be bulk billed than as a dose-response relationship. For the 2023 tripling, Craig et al. (2026) find larger effects in more affluent LGAs, but the difference disappears once the pre-reform bulk-billing rate is controlled for. For the 2025 reform, the composition of the bulk-billed pool predicts the response and the overall rate does not.

Volume. Total attendances are derived from a rounded rate, and estimates of the effect on volume fail a placebo test: seniors show a 5.5% increase relative to children, too large to attribute to their partial exposure, and the working-age estimate is +0.9% against seniors but +6.3% against children. I therefore make no claim about attendance volume, and the cost accounting assumes total attendances did not change.

5.4 Spillovers to children

The model of Craig et al. (2026) (their Proposition 1) predicts that when the incentive rises for one group, a time-constrained GP's shadow value of time rises, so groups whose incentive did not change become weakly less likely to be bulk billed. In 2025 the return

to bulk billing working-age adults rose while children's incentive did not, so the model predicts a negative spillover onto children. A negative spillover would bias the comparison with children upward: the estimate would include the fall in children's bulk billing as well as the rise for working-age adults. The data cannot separate this from the practice loading, which pushes children's bulk billing up. The net change for children was positive: their national rate rose from 90.0% to 91.3% (Table 1), and Craig et al. (2026) estimate a total effect of +0.42 points for 0–15 year olds. The comparison with seniors, who are subject to the same spillover but were partly treated, gives a similar 6.16 points. As a rough bound, if the spillover onto children were as large as the entire 0–15 effect estimated by Craig et al. (2026), the preferred estimate would fall to about 5.9 points and the cost per additional attendance would rise from $187 to about $199.

5.5 Remoteness

Table 8 reports, for all ages combined, the change in the bulk-billing rate of GP non-referred attendances between April–June 2025 and April–June 2026 by Modified Monash zone. Rates rose in every zone, by between 3.3 points (MM7) and 6.8 points (MM2). Because the rate cannot exceed 100%, a more comparable measure is the share of the remaining gap closed: about a fifth in metropolitan areas (MM1) and about a third in every other zone. This is consistent with larger dollar incentives outside the cities, but the zone data have no age split and no comparison group, so these differences are descriptive.

6 The cost of an additional bulk-billed attendance

6.1 Accounting framework

Consider the working-age group over the post-reform period. Let N be total GP attendances, held fixed, β_0 the counterfactual bulk-billing rate, Δ the effect of the reform on the rate, and b_1 the incentive paid per bulk-billed attendance after the reform, when every attendance is eligible. Before the reform only some bulk-billed attendances attracted the incentive; let e be that share, so that the incentive paid per bulk-billed attendance before the reform was $r_0 = e\, b_1$ at post-reform amounts. Actual incentive payments are $b_1(\beta_0 + \Delta)N$. Without the reform, the $\beta_0 N$ attendances that would have been bulk billed anyway would have attracted $r_0\beta_0 N$. The reform added ΔN bulk-billed attendances, so the incentive cost per additional bulk-billed attendance is

$$C \;=\; \frac{b_1(\beta_0 + \Delta) - e\, b_1\beta_0}{\Delta} \;=\; b_1\left[1 + (1-e)\,\frac{\beta_0}{\Delta}\right], \tag{3}$$

and the share of the additional spending that went to attendances that would have been bulk billed anyway is

$$S \;=\; \frac{(1-e)\,\beta_0}{(1-e)\,\beta_0 + \Delta}. \tag{4}$$

The first term in equation (3) is the incentive paid on the additional attendance itself. The second is the new incentive spread over the attendances that would have been bulk billed anyway, scaled by how many of them there are per additional attendance (β_0/Δ) and by the share that were not already attracting the incentive ($1-e$). Two features follow. Cost rises steeply as the effect falls or as baseline compliance rises, which is the core of the argument in Craig et al. (2026). And for given b_1, β_0 and Δ, assuming no prior eligibility ($e = 0$) overstates the cost by $e\,b_1\beta_0/\Delta$.

All four inputs are observed or estimated here. In the eight post-reform months $b_1 =$ \$22.73; the counterfactual rate is the observed rate less the estimated effect, $\beta_0 = 0.706$; and the pre-reform incentive per bulk-billed working-age attendance, measured over the twelve months to October 2025, was $r_0 =$ \$7.96, so $e = 0.35$: about a third of working-age bulk-billed attendances already attracted the incentive before the reform, mostly those of concession card holders.[2]

6.2 Estimates

Table 4 reports the cost. At the preferred estimate of 6.33 points, the reform converted about 3.8 million working-age attendances to bulk billing over eight months. Incentive payments for working-age adults were \$1,041 million, about \$706 million more than the attendances that would have been bulk billed anyway would have attracted at pre-reform rates. Equations (3) and (4) give a cost of \$187 per additional bulk-billed attendance and a share of 88% going to attendances that would have been bulk billed anyway. Across the confidence interval the cost ranges from \$171 to \$207. Setting $e = 0$, as if no working-age patient had been eligible before, gives \$276 at the central estimate (last column of Table 4). Prior eligibility therefore accounts for \$89 of the difference, and ignoring it raises the cost by about half.

Figure 3 plots equation (3) against the effect size for $e = 0.35$ and $e = 0$. The gap between the curves is largest when the effect is small, which is when the cost figure matters most for policy.

The practice loading. These figures exclude the BBPIP, because payments under it are not reported by age group. The MBS benefit per bulk-billed working-age attendance, excluding the incentive, was about \$59 after the reform, so the 12.5% loading is worth

[2]Measuring e as r_0/b_1 values pre-reform eligible attendances at post-reform amounts. The July 2025 indexation means pre-reform amounts were slightly lower, so e is if anything slightly understated and C slightly overstated.

about $7.40 per attendance at a qualifying practice. If every bulk-billed working-age attendance were at a qualifying practice and the whole loading were attributed to the working-age response, it would add up to $90 per additional bulk-billed attendance. The true figure is lower, because not all practices qualify and the loading also induces bulk billing of other groups, but it is not zero, so $187 is a lower bound on the full cost of the package per additional working-age bulk-billed attendance.

7 Comparison with Craig, Findlay and Lilley (2026)

Table 5 sets the two papers' estimates of the 2025 reform side by side. Craig et al. (2026) use monthly data by local government area and age group, estimate a regression-discontinuity-in-time model for each age group in logit form, with a linear trend and month-of-year effects, and use the 0–15 response to separate the effect of eligibility from the effect of the practice loading. Both papers now use the same LGA data, so the differences between them come from design and accounting. The design here differences the working-age response against children within each LGA and month, which removes any national shock common to both groups at the cost of not identifying the practice loading's own effect, and of the spillover concern discussed in Section 5.4.

Effect. Their total effect for 16–64 year olds is 6.56 points and for 0–15 year olds 0.42 points, a difference of 6.14 points, against 6.33 points here. Their direct (eligibility) effect for 16–64 year olds, 5.93 points, is within my confidence interval. Both papers find a smaller positive effect for seniors.

Cost. The difference in the cost figures is mostly accounting. Equation (3) with their inputs (their direct effect of 5.93 points, their average incentive of $25.23 and $e = 0$) and the counterfactual rate from this paper gives $326, close to their $322. Starting from their inputs and changing one at a time, using the observed incentive per bulk-billed attendance instead of the average schedule amount lowers the cost from $326 to $293, netting out prior eligibility lowers it to $199, and using the effect estimated here lowers it to $187. Prior eligibility accounts for $89 to $105 of the difference, depending on the order in which the adjustments are made, and is the largest single component. Both papers agree on the substantive conclusion: the large majority of the reform's incentive spending is a transfer for attendances that would have been bulk billed anyway.

8 The November 2023 tripling

For the 2023 tripling the roles reverse: children and seniors (mostly eligible) are treated, and working-age adults (mostly ineligible) are the comparison group. Craig et al. (2026)

estimate an effect of 3 points using a six-month reference period, chosen because trends were plausibly parallel over that window. Figure 4 shows the longer history in the state-level data. The gap between children's and working-age adults' bulk-billing rates was stable at about 11 points in 2018 and 2019, fell during the COVID-19 period, and then widened from 6.9 points in July 2021 to 18.0 points in July 2023, before the tripling. Craig et al. (2026) read the flat six months as the gap stabilising after the pandemic. By mid-2023, however, the gap was well above its 2018–19 level, so the widening was not simply a recovery from the pandemic that had run its course, and the data before their window give no indication that it was about to stop.

Table 6 estimates the 2023 effect on the LGA panel with pre-periods of different lengths, in percentage points and in log-odds, with LGA-by-month effects and weighting as in the main analysis. With a six-month pre-period the estimate is 2.8 points, essentially the estimate of Craig et al. (2026), and the log-odds pre-trend over that window is small and slightly negative, although in percentage points the gap was still widening by 0.26 points a month. Lengthening the pre-period raises the estimate to 3.6 points (twelve months), 7.6 points (from July 2021) and 9.7 points (from January 2018), and in every window other than the shortest the pre-trend is positive in both scales: the gap was already widening. The immediate change in the gap at November 2023 was only 1.2 points relative to children and 1.7 points relative to seniors. The comparison group was also partly treated: working-age concession card holders were eligible for the tripled incentive, and incentive payments for working-age adults rose from $13.6 million in October 2023 to $38.1 million in November 2023.

Three features of the six-month window deserve note. First, it was selected because trends over it appeared parallel. Conditioning on a passed pre-test distorts both the estimate and its confidence interval when the test has low power (Roth, 2022), as a test on five pre-period months does. Second, the tripling was announced in the May 2023 Budget, so the whole reference period follows the announcement, and Craig et al. (2026) note a jump in bulk billing in the announcement month. Practices that expected the tripled incentive may have changed their billing before November, in which case the reference period does not represent untreated behaviour. Third, whether trends are parallel depends on the scale. Over the six-month window they are close to parallel in log-odds but not in percentage points, and when two groups start from different levels, parallel trends in one scale generally implies non-parallel trends in the other (Roth and Sant'Anna, 2023).

Table 7 applies the sensitivity analysis of Rambachan and Roth (2023) to an event study on the LGA panel, with October 2023 as the reference month and the post-period running to October 2025. Under the smoothness restriction, the slope of the differential trend may change by at most M per month after the tripling, so $M = 0$ extends the pre-period trend linearly. Under the relative magnitudes restriction, the month-to-month

change in the differential trend after the tripling is at most $\bar{M}$ times the largest such change in the pre-period. In log-odds, the scale used by Craig et al. (2026), the average effect over the two years after the tripling is 0.245, close to their estimate. It survives linear extrapolation of the pre-trend but not a change in slope of 0.002 log-odds per month. In the data, the slope fell from 0.023 per month between July 2022 and April 2023 to about zero between May and October 2023, a change of about 0.004 per month. Under relative magnitudes the average effect is no longer significant once post-period changes may reach 0.6 times the largest change within the six-month window. In percentage points, extending the six-month pre-trend reverses its sign. The November 2023 effect, by contrast, survives much larger departures from parallel trends in both scales, because a smooth trend cannot produce a one-month jump of that size.

The positive effect is robust; its size is not. A six-month window during which trends happen to look parallel, after two years of divergence, is a fragile basis for the magnitude. The cost per additional bulk-billed attendance that Craig et al. (2026) report for the 2023 tripling ($479) is computed from the 3-point effect and inherits this uncertainty. Person-level data that observe concession card status would allow a comparison between eligible and ineligible patients of the same age, which removes the need for an age-group contrast altogether.

9 Discussion and conclusion

Extending the bulk-billing incentive to all patients in November 2025 raised the bulk-billing rate of working-age adults by about 6.3 percentage points relative to children within eight months, an effect that was immediate and still growing, and that agrees closely with Craig et al. (2026). The incentive cost of each additional bulk-billed attendance was about $187 before the practice loading, and 88% of the additional incentive spending went to attendances that would have been bulk billed anyway. The reform was expensive per additional bulk-billed attendance, but less so than an accounting that ignores prior eligibility suggests. The framework in Section 6 shows why: the cost depends on the share of attendances already receiving the incentive as well as on baseline compliance and the size of the response.

Several limits apply. The estimate is a reduced-form effect of eligibility for working-age adults, net of the practice loading's common effect and of any spillover onto children. The policy is national, so a coincident shock specific to working-age adults cannot be ruled out. The placebo dates do not rule one out either: against children they are significantly negative, which points to drift against finding an effect rather than towards one, and that drift had largely flattened before the reform. Total attendances are derived from a rounded rate, so I make no claim about volume, and the data contain no out-of-pocket costs. With eight months of post-reform data, persistence is unknown.

Most of these limits stem from aggregation. Linked person-level data, such as MBS claims linked to concession card status and income in the Person Level Integrated Data Asset, would allow the reform to be evaluated by comparing newly eligible and previously eligible adults of the same age, would make the 2023 tripling identifiable without an age-group contrast and its two-year effect estimable without assumptions about how the age-group trends would have evolved, would allow direct measurement of attendance volume and spillovers, and would allow practice-level responses to the 12.5% loading to be studied. These are natural next steps.

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Tables

Table 1: Summary statistics, national, by age group

	Children (0–15)		Working age (16–64)		Seniors (65+)	
	Pre	Post	Pre	Post	Pre	Post
Bulk-billing rate (%)	90.0	91.3	69.1	76.9	87.3	89.1
GP attendances (m/month)	1.58	1.46	7.50	7.44	4.78	4.70
Bulk-billed attendances (m/month)	1.43	1.33	5.19	5.72	4.17	4.19
Incentive benefits ($m/month)	32.1	31.2	40.2	130.1	74.0	95.4
Incentive per bulk-billed attendance ($)[a]	22.93	23.46	7.96	22.73	18.22	22.79

Notes: Pre is November 2023 to October 2025 (24 months); post is November 2025 to June 2026 (8 months). GP attendances are derived as bulk-billed attendances divided by the bulk-billing rate. [a] Pre-period incentive per bulk-billed attendance uses November 2024 to October 2025. Source: Australian Institute of Health and Welfare (2026).

Table 2: Effect of the November 2025 reform on the working-age bulk-billing rate (percentage points)

	(1)	(2)	(3)	(4)
Comparison group	Children	Seniors	Children & seniors	Children
Fixed effects	LGA × month	LGA × month	LGA × month	Month
Working-age × post Nov 2025	6.33***	6.16***	6.20***	6.40***
SE, clustered by LGA	(0.31)	(0.23)	(0.25)	(0.30)
SE, clustered by state	[0.65]	[0.45]	[0.50]	[0.58]
95% CI, LGA clusters	[5.7, 6.9]	[5.7, 6.6]	[5.7, 6.7]	[5.8, 7.0]
95% CI, state clusters, $t(7)$	[4.8, 7.9]	[5.1, 7.2]	[5.0, 7.4]	[5.0, 7.8]
Pre-reform working-age rate (%)	69.1	69.1	69.1	69.1
LGAs	393	393	393	393
Observations	25,152	25,152	37,728	25,152

Notes: Estimates of equation (1) on the LGA × age group × month panel, November 2023 to June 2026, weighted by mean monthly GP attendances of each cell before the reform. All columns include LGA-by-age-group fixed effects; columns 1 to 3 add LGA-by-month fixed effects and column 4 month fixed effects. Standard errors clustered by LGA in parentheses and by state in brackets. Stars refer to LGA-clustered standard errors. Column 1 is the preferred specification. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 3: Robustness checks

Check	Estimate	SE	p
Panel A: specification			
Leave one state out (min–max)	6.01 (Vic out) to 7.08 (NSW out)		
Excluding unincorporated areas	+6.42	0.31	0.000
Unweighted (average across LGAs)	+8.12	0.29	0.000
Log-odds of the bulk-billing rate	+0.296	0.02	0.000
Panel B: placebo tests			
Placebo group: seniors vs children	+0.27	0.09	0.004
Placebo date May 2024, vs children	-0.77	0.09	0.000
Placebo date Nov 2024, vs children	-0.35	0.07	0.000
Placebo date May 2025, vs children	-0.17	0.09	0.051
Placebo date May 2024, vs children & seniors	-0.03	0.07	0.674
Placebo date Nov 2024, vs children & seniors	+0.11	0.07	0.099
Placebo date May 2025, vs children & seniors	+0.46	0.07	0.000
Panel C: differential linear pre-trends (pp per month)			
Working-age vs children	-0.034	0.006	0.000
Working-age vs seniors	+0.027	0.006	0.000
Working-age vs children & seniors	+0.012	0.005	0.029

Notes: Specifications as in Table 2, column 1 (working-age adults against children, LGA-by-age-group and LGA-by-month fixed effects, weighted, standard errors clustered by LGA), unless stated. Placebo dates and pre-trends use the pre-reform months only (November 2023 to October 2025). The placebo group compares seniors with children around November 2025.

Table 4: Incentive cost of additional bulk-billed working-age attendances, November 2025 to June 2026

Scenario	Effect (pp)	Additional bulk-billed visits (m)	Extra incentive spend ($m)	Cost per additional visit ($)	Share on visits bulk-billed anyway	Cost if no prior eligibility ($)
Low	5.7	3.4	703	207	89%	306
Central	6.3	3.8	706	187	88%	276
High	6.9	4.1	709	171	87%	252

Notes: Effects from Table 2, column 1; low and high use the bounds of the 95% confidence interval with LGA clusters. Attendances, incentive payments and r_0 are national totals from the state table. Additional visits are the effect times total working-age attendances (59.5 million), assuming total attendances did not change. Extra spend is actual incentive benefits ($1,041 million) less the pre-reform incentive per bulk-billed attendance ($7.96) times attendances that would have been bulk billed anyway. Cost per additional visit and the share are equations (3) and (4) with $e = 0.35$; the last column sets $e = 0$. Excludes BBPIP payments and spending on seniors.

Table 5: Estimates of the November 2025 reform: Craig et al. (2026) and this paper

	Craig et al. (2026)	This paper
Data	AIHW monthly, local government area × age group, to March 2026	AIHW monthly, local government area × age group, to June 2026
Design	Regression discontinuity in time by age group; logit; LGA fixed effects, month-of-year effects, linear trend	Difference-in-differences, working-age vs children; LGA-by-age-group and LGA-by-month fixed effects
Weighting	Appointments	GP attendances
Inference	Clustered by LGA	Clustered by LGA (393); by state reported
Effect, 16–64 (pp)	6.56 total; 5.93 direct	–
Effect, 0–15 (pp)	0.42	–
Effect, 65+ (pp)	1.18	–
16–64 relative to 0–15 (pp)	6.14 (implied)	6.33 [5.7, 6.9]
65+ relative to 0–15 (pp)	0.76 (implied)	0.27
Prior eligibility of 16–64	Assumed none	Observed: $e = 0.35$
Cost per additional bulk-billed attendance	\$322	\$187 (\$276 with $e = 0$)
Out-of-pocket costs, 16–64	−21.5%	Not available
2023 tripling	+3 pp (six-month reference period)	Jump robust; size sensitive to pre-period (2.8 to 9.7 pp) and to trend (Table 7)

Notes: Craig et al. (2026) figures from their Tables 8 to 10 and text. Implied differences are computed from their reported totals by age group.

Table 6: The November 2023 tripling: sensitivity to the pre-period

	Percentage points		Log-odds		
Pre-period	DiD	Pre-trend (per month)	DiD	Pre-trend (per month)	LGAs
6 months	2.77 (0.14)	+0.256 (0.025)	0.252 (0.011)	-0.0020 (0.0015)	398
12 months	3.59 (0.16)	+0.324 (0.021)	0.273 (0.012)	+0.0100 (0.0014)	395
From July 2021	7.62 (0.28)	+0.482 (0.020)	0.378 (0.017)	+0.0138 (0.0012)	393
From January 2018	9.70 (0.37)	+0.088 (0.006)	0.418 (0.023)	+0.0018 (0.0004)	393

Notes: Difference-in-differences of the bulk-billing rate of children and seniors (pooled) against working-age adults around 1 November 2023, sample ending October 2025, on the LGA panel with LGA-by-age-group and LGA-by-month fixed effects, weighted by pre-period attendances. Each window keeps the LGAs with all three age groups reported in every month. The pre-period starts at the date shown. Pre-trends are differential linear trends estimated on the pre-period only. Standard errors clustered by LGA in parentheses.

Table 7: The November 2023 tripling: sensitivity to violations of parallel trends

Restriction	Pre-period	M or $\bar{M}$	95% confidence set
A. Average effect over 24 months, log-odds (estimate 0.245)			
Conventional	–		[0.225, 0.266]
Smoothness, M	6 months	0	[0.196, 0.276]
Smoothness, M	6 months	0.001	[0.026, 0.343]
Smoothness, M	6 months	0.002	[−0.127, 0.427]
Smoothness, M	From July 2022	0	[0.140, 0.208]
Smoothness, M	From July 2022	0.002	[−0.115, 0.438]
Relative magnitudes, $\bar{M}$	6 months	0.5	[0.047, 0.443]
Relative magnitudes, $\bar{M}$	6 months	0.6	[0.007, 0.478]
Relative magnitudes, $\bar{M}$	6 months	0.75	[−0.048, 0.539]
B. Average effect over 24 months, percentage points (estimate 1.94)			
Conventional	–		[1.71, 2.18]
Smoothness, M	6 months	0	[−2.18, −0.89]
Smoothness, M	From July 2022	0	[−1.93, −0.98]
C. November 2023 effect, log-odds (estimate 0.159)			
Conventional	–		[0.143, 0.175]
Smoothness, M	6 months	0.05	[0.082, 0.213]
Relative magnitudes, $\bar{M}$	6 months	3	[0.066, 0.257]
D. November 2023 effect, percentage points (estimate 1.46)			
Conventional	–		[1.26, 1.66]
Smoothness, M	6 months	0.5	[0.12, 1.51]
Smoothness, M	6 months	1	[−0.38, 2.01]

Notes: Robust 95% confidence sets of Rambachan and Roth (2023), computed with the HonestDiD package (version 0.2.8). The event study regresses the outcome on interactions of the treated group (children and seniors, pooled) with each month from January 2018 to October 2025, against working-age adults, with LGA-by-age-group and LGA-by-month fixed effects, weighted by pre-period attendances, standard errors clustered by LGA; October 2023 is the reference month. Panels A and B target the average of the 24 post-period coefficients, Panels C and D the November 2023 coefficient. Because the estimates are measured relative to October 2023, they differ from the static estimates in Table 6. Smoothness: the slope of the differential trend changes by at most M per month. Relative magnitudes: post-period month-to-month changes are at most $\bar{M}$ times the largest pre-period change. Pre-period: months used to bound the trend.

Table 8: Incentive amounts and change in bulk billing by Modified Monash zone, all ages

Zone	Standard incentive ($)	Tripled incentive ($)	Rate Apr–Jun 2025 (%)	Change to Apr–Jun 2026 (pp)	Share of remaining gap closed
MM1 Metropolitan	7.35	21.85	78.5	+4.3	20%
MM2 Regional centres	11.05	33.25	78.2	+6.8	31%
MM3 Large rural towns	11.70	35.30	82.1	+6.3	35%
MM4 Medium rural towns	11.70	35.30	83.3	+5.4	32%
MM5 Small rural towns	12.50	37.55	82.9	+6.4	37%
MM6 Remote	13.15	39.65	79.8	+6.6	33%
MM7 Very remote	14.05	42.05	90.4	+3.3	35%

Notes: Incentive amounts are benefits per eligible service in the July 2025 schedule (Department of Health, Disability and Ageing, 2025); zones 3 and 4 share one item. Bulk-billing rates are for GP non-referred attendances, all ages (Department of Health, Disability and Ageing, 2026). Share of remaining gap closed is the change divided by (100 minus the April–June 2025 rate). Descriptive only.

Figures

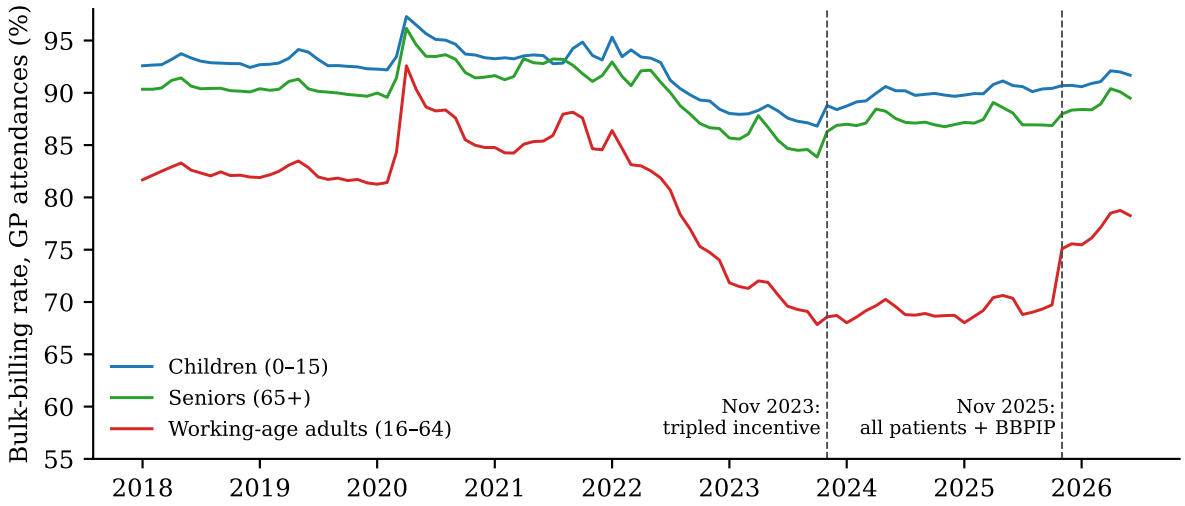


Figure 1: National bulk-billing rate of GP attendances by age group, January 2018 to June 2026

Notes: National rates are bulk-billed attendances divided by derived total attendances, summed over states. Source: Australian Institute of Health and Welfare (2026).

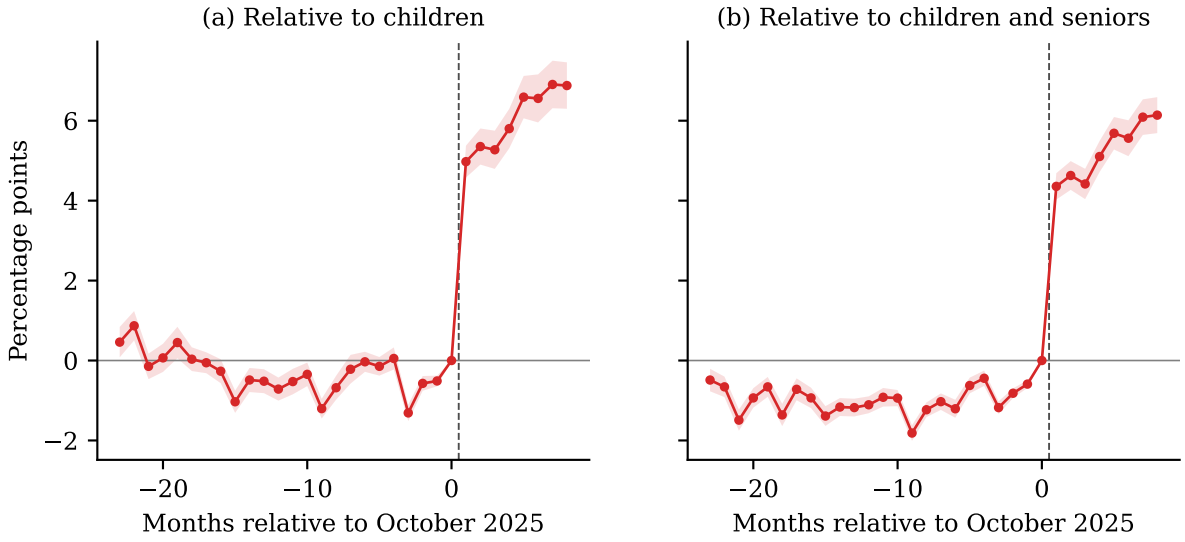


Figure 2: Event study: working-age bulk-billing rate relative to children and to children and seniors

Notes: Coefficients β_k from equation (2), relative to October 2025, on the LGA panel, weighted. Panel (a) is the preferred comparison with children. Shaded areas are pointwise 95% intervals with standard errors clustered by LGA.

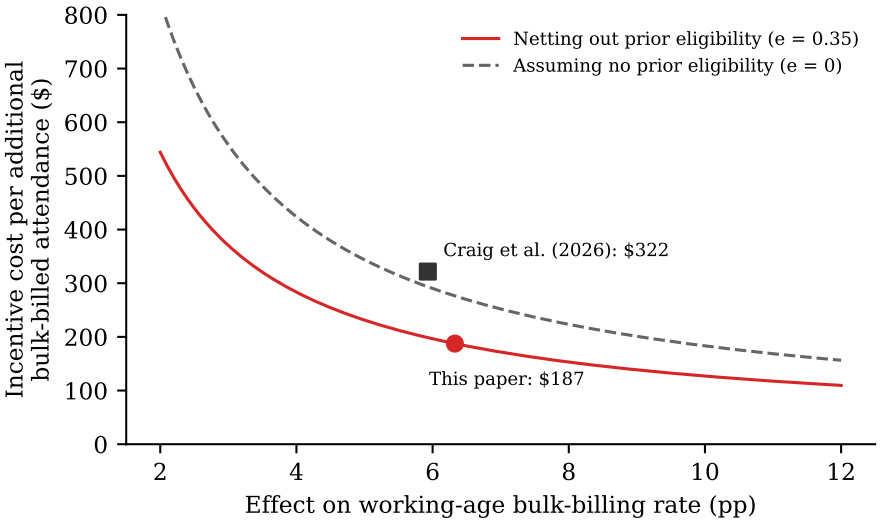


Figure 3: Incentive cost per additional bulk-billed attendance by effect size

Notes: Equation (3) with $b_1 = \$22.73$ and $\beta_0 = 0.706$. The solid line nets out prior eligibility ($e = 0.35$); the dashed line assumes none ($e = 0$). The square marks the estimate of Craig et al. (2026), which uses their direct effect and average incentive.

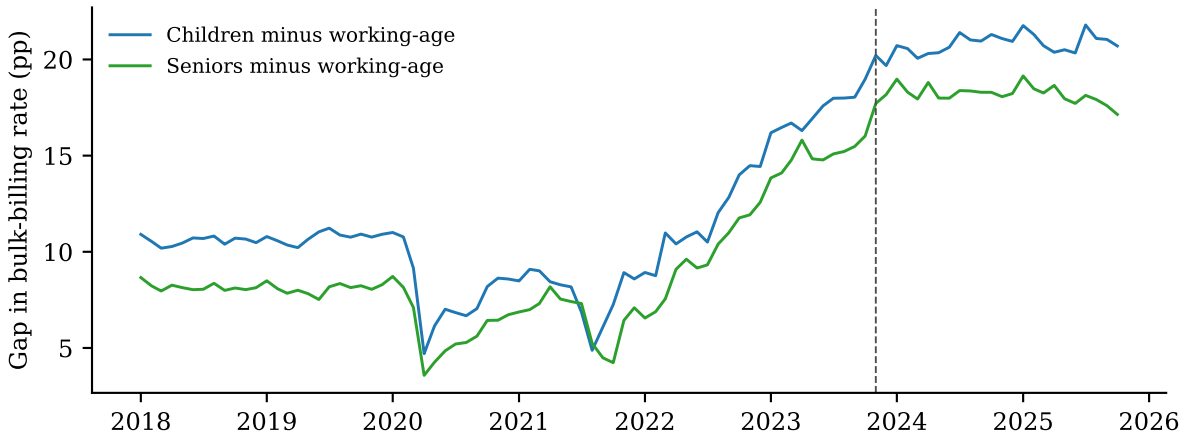


Figure 4: Gap between the bulk-billing rates of children or seniors and working-age adults, January 2018 to October 2025

Notes: The dashed line marks the November 2023 tripling. Source: Australian Institute of Health and Welfare (2026).

A Additional results

State-level estimates. Table 9 reports equation (1) estimated on the state × age group × month panel from supplementary Table 2 (8 states × 3 age groups × 32 months), with state-by-age-group and month fixed effects. Inference uses the restricted wild cluster bootstrap with the null imposed, Webb weights and 9,999 replications, with confidence intervals by test inversion. The estimates, 6.43 points against children, 6.05 against seniors and 6.15 pooled, are within 0.2 points of the LGA estimates, with wider intervals. Unweighted state estimates (Table 10) are larger, 7.6 to 8.0 points, because the smallest jurisdictions had the largest responses: the Northern Territory, with under 1% of working-age attendances, had a response of about 18 points. Table 11 reports the cost on alternative state-level bases.

Table 9: State-level estimates (percentage points)

Comparison group	(1) Children	(2) Seniors	(3) Children & seniors
Working-age × post Nov 2025	6.43**	6.05**	6.15**
	(0.63)	(0.62)	(0.62)
Wild cluster bootstrap p	0.014	0.014	0.013
95% CI (test inversion)	[5.2, 9.9]	[4.8, 8.7]	[4.9, 9.1]
Pre-reform working-age rate (%)	69.1	69.1	69.1
Observations	512	512	768

Notes: Estimates of equation (1) on the state × age group × month panel, November 2023 to June 2026, weighted by mean monthly GP attendances before the reform, with state-by-age-group and month fixed effects. CR1 standard errors clustered by state (8 clusters) in parentheses. p-values from the restricted wild cluster bootstrap (Webb weights, 9,999 replications); confidence intervals by inverting the bootstrap test. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 10: Unweighted state-level estimates (percentage points)

Comparison group	(1) Children	(2) Seniors	(3) Children & seniors
Working-age × post Nov 2025	7.96***	7.63***	7.80***
	(2.05)	(1.46)	(1.72)
Wild cluster bootstrap p	0.007	0.003	0.004
95% CI (test inversion)	[3.9, 12.6]	[4.8, 10.9]	[4.4, 11.7]
Pre-reform working-age rate (%)	63.0	63.0	63.0
Observations	512	512	768

Notes: As Table 9 but unweighted. The pre-reform rate is the simple mean of state rates.

Table 11: Incentive cost on alternative state-level bases

Scenario	Effect (pp)	Additional bulk-billed visits (m)	Extra incentive spend ($m)	Cost per additional visit ($)	Share on visits bulk-billed anyway	Cost if no prior eligibility ($)
Weighted, pooled						
Low	4.9	2.9	699	242	91%	360
Central	6.1	3.7	705	193	88%	284
High	9.1	5.4	719	133	83%	193
Unweighted, pooled						
Low	4.4	2.6	697	269	92%	401
Central	7.8	4.6	713	154	85%	224
High	11.7	7.0	732	105	78%	149

Notes: As Table 4, using the state-level estimates in Tables 9 and 10 for the pooled comparison group, weighted and unweighted, with wild bootstrap confidence intervals.

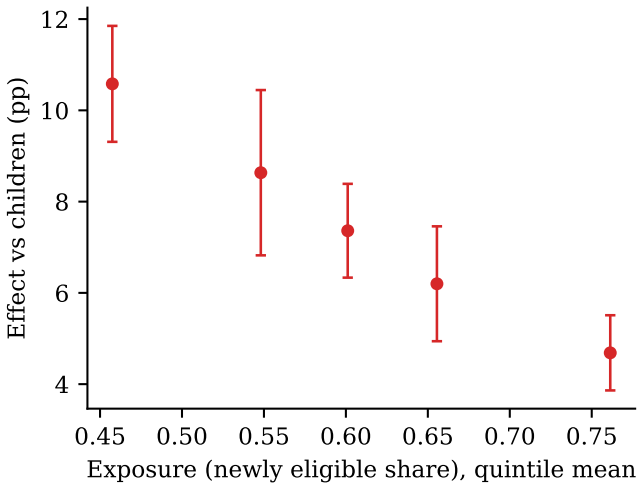


Figure 5: Working-age response by exposure to the reform, LGA quintiles

Notes: Exposure is one minus the ratio of pre-reform (November 2024 to October 2025) to post-reform incentive paid per bulk-billed working-age attendance in each LGA. LGAs are grouped into fifths by exposure; each point is the estimate of Table 2, column 1, within that group, plotted at the group's attendance-weighted mean exposure, with a 95% interval (standard errors clustered by LGA).